

Assessment Task 1 - Regression Report

Predicting ORCL Next-Day Returns: A Regression Comparison of Linear
Regression and Random Forest

Student(s) Name(s):

Santiago Cruz Lopez | **ID:** 2584023

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Professor Vahid Rafe

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Wolverhampton, UK

The University of Wolverhampton

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1. Introduction

This report focuses on Task 1 of the portfolio, which involves building and comparing two regression algorithms using a real-world dataset. The specific problem addressed is predicting the percentage return of Oracle Corporation (ORCL) common stock for the next trading day, based on that day's technical indicators. This is a challenging, realistic regression problem, as public information alone often makes financial returns nearly unpredictable. This idea is formalized in the Efficient Market Hypothesis (Fama, 1970). This task evaluates two different regression algorithms in light of this hypothesis, rather than simply accepting it. Each algorithm is implemented in its own separate notebook, allowing for independent verification of results before making a comparison here.

2. Dataset and Method

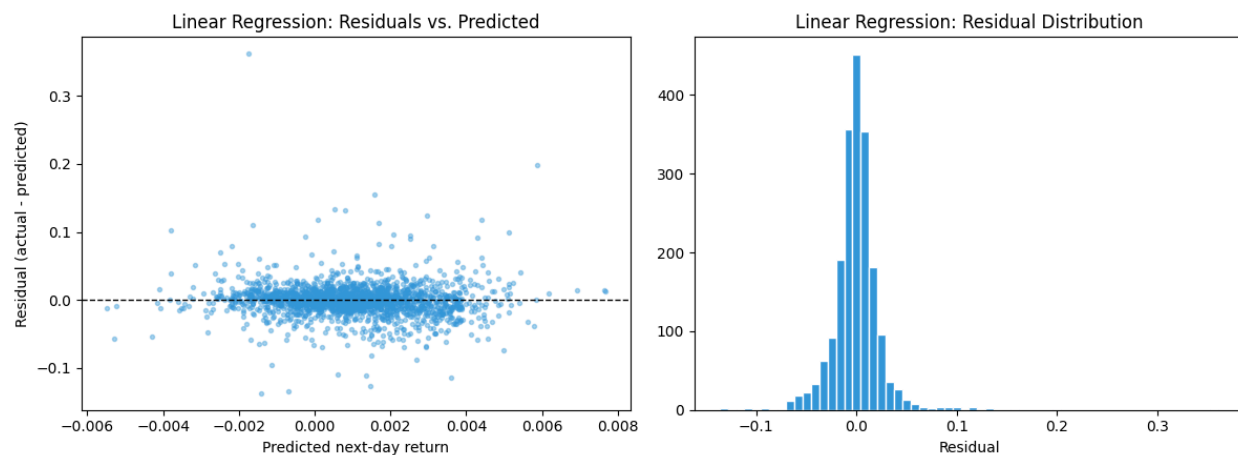
The dataset consists of Oracle Corporation's (ORCL) daily open, high, low, and close price history, along with pre-computed technical indicators such as moving averages, RSI (Relative Strength Index), and TSI (Trend Strength Indicator), spanning from 1986 to 2026, totaling 10,185 rows (TradingView, 2026).

Eight features were engineered from each day's indicators, including daily return, 10-day volatility, percentage distance from three moving averages, and the RSI and TSI oscillators. The target variable is the percentage return for the following day, which makes this a regression task rather than a classification task (Hyndman & Athanasopoulos, 2021).

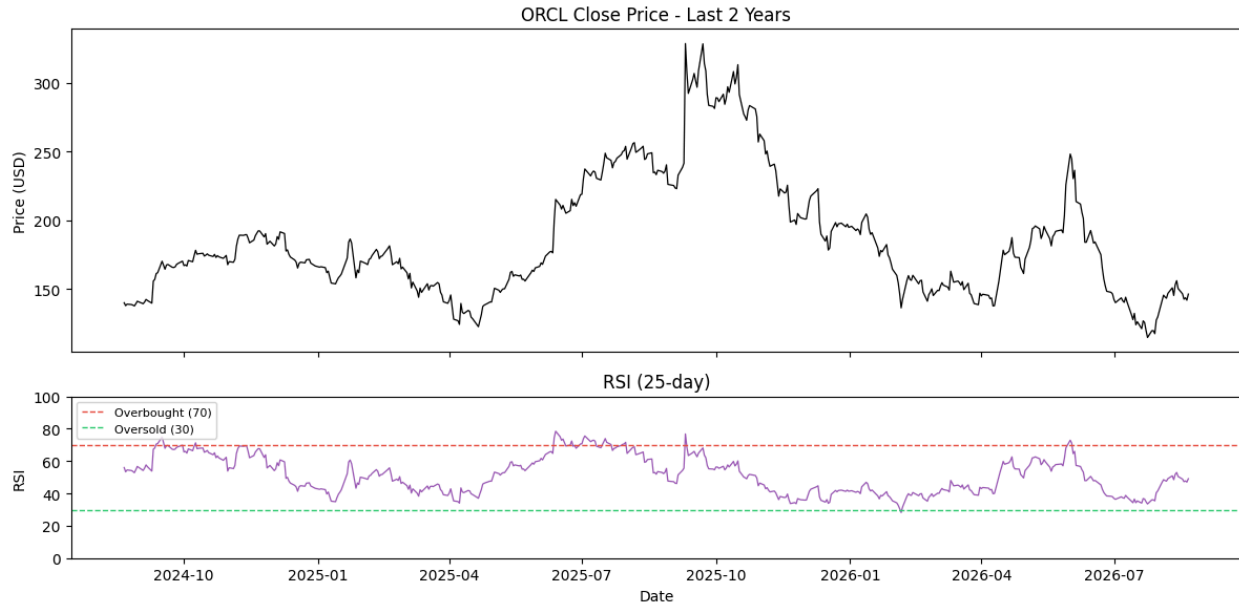
The data was split chronologically (not randomly): the first 80% of the dates were allocated for training, while the last 20% were used for testing. This approach avoids using future information for training the model relative to any test day. Both notebooks follow this identical preprocessing pipeline intentionally, ensuring that any differences in the results arise solely from the choice of algorithm, rather than inconsistencies in data handling (Hyndman & Athanasopoulos, 2021).

3. Algorithm 1: Linear Regression

Linear regression models the target variable as a weighted sum of the input features, plus a constant. The equation can be expressed as $\hat{y} = b_0 + b_1x_1 + \dots + b_8x_8$. The weights are chosen to minimize the sum of squared errors between the predictions and the actual training values, following the "ordinary least squares" criterion (James et al., 2023). We selected this approach as an interpretable baseline because each coefficient clearly indicates how much the predicted return changes with a one-unit change in that feature, while keeping the other features constant. This transparency makes the model's reasoning fully auditable instead of a black box.

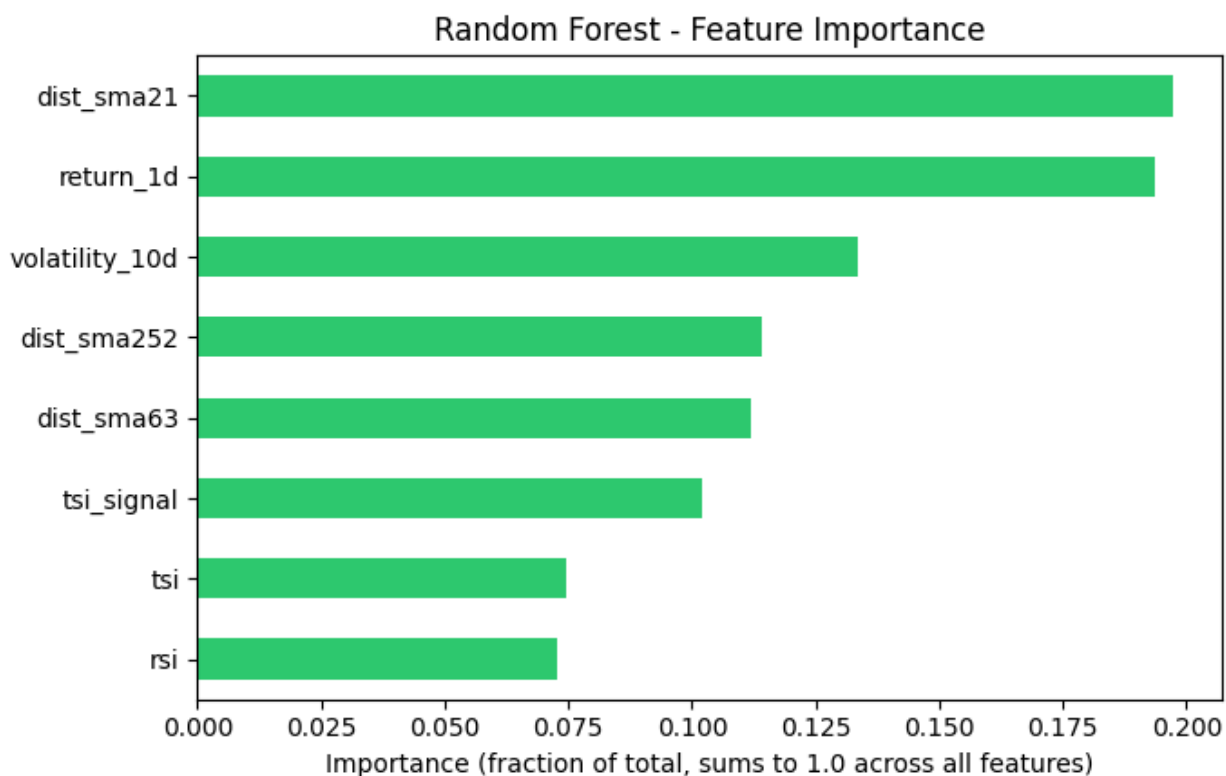


To concretely illustrate the model's coefficients, the notebook includes a plot of the raw RSI (Relative Strength Index) indicator alongside price data from the last two years. It also features overbought and oversold reference lines. Throughout this period, the RSI frequently moves between these two zones. However, the model assigns it a coefficient of nearly zero, providing visual confirmation that these swings do not have any linear predictive value for the next day's return.

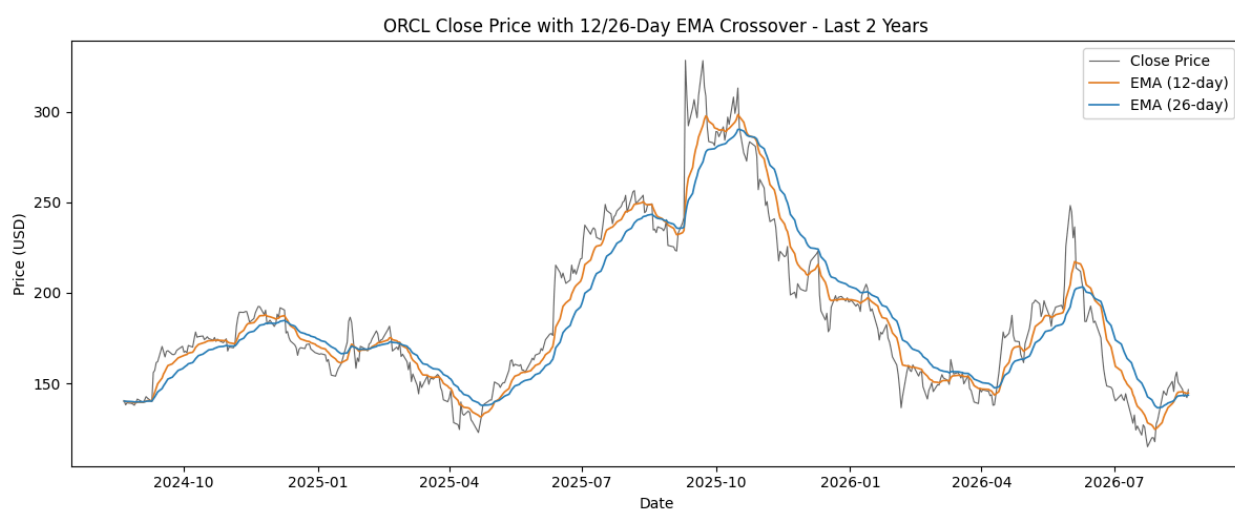


4. Algorithm 2: Random Forest Regressor

A Random Forest is an ensemble of numerous decision trees (200 were used in this case), each trained on a random bootstrap sample of the training data and considering only a random subset of features at each split. The predictions from these trees are then averaged (Breiman, 2001). This "bagging" approach enables the model to capture non-linear relationships and interactions between features. For instance, high volatility combined with an extreme RSI behaves differently than either factor alone, something that a Linear Regression model's straight-line assumption cannot accommodate. To prevent overfitting—given that there are only eight features and a particularly noisy target—the tree depth was intentionally limited (`max_depth=5`, `min_samples_leaf=10`). An unconstrained forest would likely fit the noise rather than the actual signal (Hastie et al., 2009).



The same notebook also calculates a 12-day/26-day Exponential Moving Average (EMA) crossover. This indicator is not included in the eight trained features since only Simple Moving Averages were available in the source data. The EMA crossover is used purely to visualize a trend signal that the model never had the chance to evaluate. This is directly relevant to Section 7 below.



5. Results and Model Comparison

Table 1 summarises both models on the held-out test set.

Model	MAE	RMSE	MAPE	Directional Accuracy	R ²
Linear Regression	0.0154	0.0246	118.5%	50.8%	−0.40%
Random Forest Regressor	0.0153	0.0246	111.0%	52.5%	−0.11%

Random Forest performs slightly better on every single-split metric; however, the 5-fold cross-validation results from each notebook reveal a more nuanced perspective. Specifically, Random Forest's mean cross-validation Mean Absolute Error (CV MAE) is marginally lower at 0.01746 compared to Linear Regression's 0.01748. Conversely, its mean cross-validation Root Mean Square Error (CV RMSE) is slightly higher at 0.02553, while Linear Regression's is 0.02547. This raises the concern that the claim regarding the single-split results may not be fully robust and verifiable by the reader.

Overall, the conclusion is that the two models are statistically indistinguishable for this problem. Random Forest's "win" in the single split is small enough to fall within normal fold-to-fold noise, indicating it is not decisively a better algorithm.

Here are the results from the 5-fold TimeSeriesSplit cross-validation:

Algorithm 1: Linear Regression

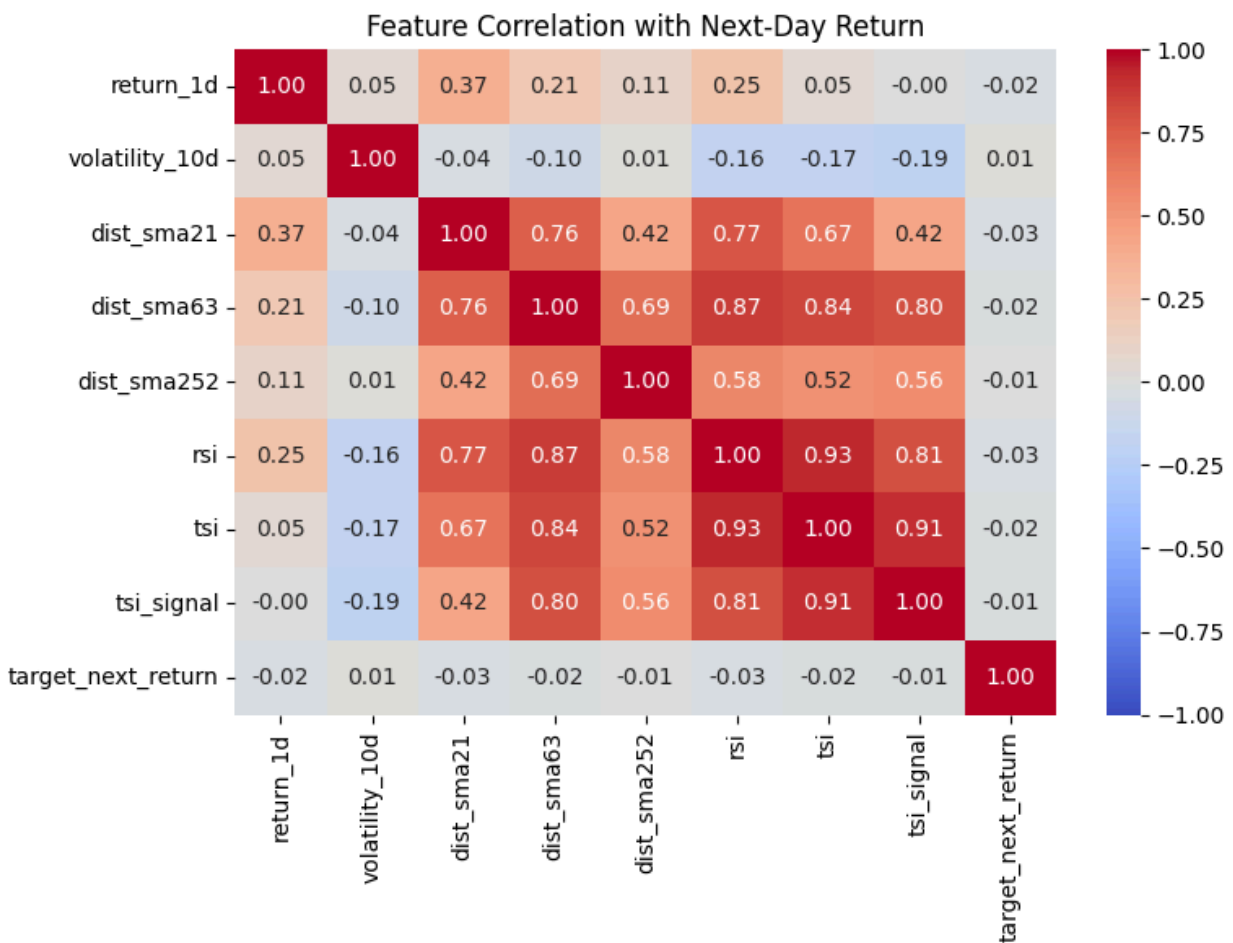
- Mean CV MAE: 0.01748 (std 0.00568)
- Mean CV RMSE: 0.02547 (std 0.00797)

Algorithm 2: Random Forest Regressor

- Mean CV MAE: 0.01746 (std 0.00573)
- Mean CV RMSE: 0.02553 (std 0.00811)

6. Performance Analysis

The marginal single-split performance of the Random Forest model, along with its slightly inferior cross-validated performance in terms of RMSE, can both be attributed to the same underlying issue: none of the eight features show a meaningful relationship with the target variable. The feature with the strongest correlation, `dist_sma21`, has a correlation coefficient of -0.031, which is close to zero. This pattern of near-zero correlation is consistent across all features. A detailed analysis of one feature is provided in Section 3 of the Linear Regression notebook, specifically in the RSI deep-dive.



When no feature carries significant signal, a flexible model like Random Forest can only "improve" upon a linear model by capturing small, non-generalizing patterns from the training noise. This is why its cross-validated RMSE does not consistently outperform the linear model, even though it may have a slight advantage in single-split MAE. Another contributing factor is

severe multicollinearity; for instance, RSI and TSI have a correlation of 0.93, while TSI and TSI_signal have a correlation of 0.91.

Table 5. Section 3.1 (identical in both notebooks)

	return_ 1d	volatility_1 0d	dist_sma 21	dist_sma 63	dist_sma2 52	rsi	tsi	tsi_sign al
return_1d	1	0.05	0.37	0.21	0.11	0.25	0.05	0
volatility_ 10d	0.05	1	-0.04	-0.1	0.01	-0.16	-0.17	-0.19
dist_sma2 1	0.37	-0.04	1	0.76	0.42	0.77	0.67	0.42
dist_sma6 3	0.21	-0.1	0.76	1	0.69	0.87	0.84	0.8
dist_sma2 52	0.11	0.01	0.42	0.69	1	0.58	0.52	0.56
rsi	0.25	-0.16	0.77	0.87	0.58	1	0.93	0.81
tsi	0.05	-0.17	0.67	0.84	0.52	0.93	1	0.91
tsi_signal	0	-0.19	0.42	0.8	0.56	0.81	0.91	1

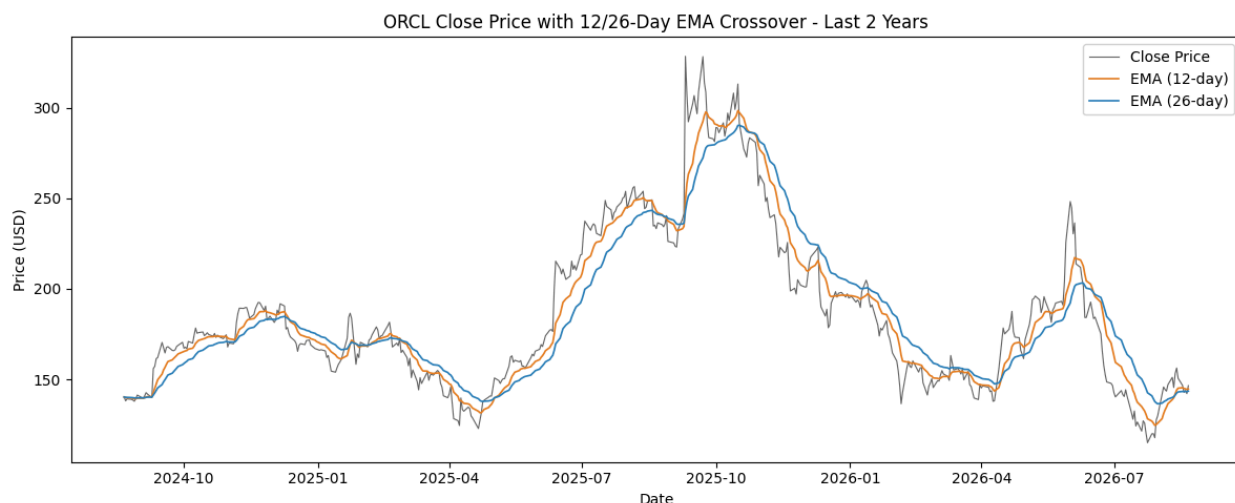
Highly correlated features do not bias the predictions of a Random Forest model because it only requires one feature from a correlated group to make a split. In contrast, correlated features can lead to unstable and difficult-to-interpret coefficients in Linear Regression, even though the overall model fit remains unaffected (Hastie et al., 2009). This presents a second, independent reason why Random Forest is a more robust choice in this scenario, despite both models ultimately being unhelpful. The actual versus predicted scatter plots for each model clearly illustrate this; both models' predictions cluster in a narrow band around zero, regardless of the true returns, and are far from the line of perfect prediction.

7. Suggested Improvements

The analysis above leads to three specific, evidence-based recommendations, rather than generic suggestions.

First, we should eliminate redundant features. The correlation table in Section 6 indicates that 'tsi' and 'tsi_signal' are nearly duplicates of 'rsi.' By retaining only one oscillator—such as 'rsi,' which has a comparable importance in the Random Forest analysis—we can simplify the model without losing any valuable information.

Second, we need to incorporate data that the current feature set lacks. Insights from the Random Forest notebook's in-depth examination of Exponential Moving Averages (EMA) in Section 7 reveal that there are no existing EMA features among the current eight. Since EMAs respond more quickly to trend changes than the Simple Moving Averages (SMA) already in use, adding an EMA-based distance feature alongside the existing SMA-based ones presents a concrete, low-cost opportunity for further experimentation, rather than a speculative idea.



There was no trading-volume column available in this export, which is a significant omission. Volume-based indicators, such as On-Balance Volume, are standard technical analysis tools that complement price-based indicators, and their absence here is noteworthy.

Additionally, it's important to reevaluate the prediction horizon and validation strategy. The current 5-fold walk-forward cross-validation, which has already been run in both notebooks,

should be expanded to include more folds, along with hyperparameter tuning for each fold. Furthermore, extending the prediction horizon to 5 or 21 days could provide more meaningful signals than focusing solely on a single day, as short-horizon returns are most closely aligned with a pure random walk (Hyndman & Athanasopoulos, 2021).

8. Conclusion

Both Linear Regression and Random Forest were implemented independently in the notebooks **2584023_Task1_Algo1.ipynb** and **2584023_Task1_Algo2.ipynb**, respectively. Both models used the same preprocessing, features, and evaluation methodology, and each model was cross-validated in isolation. While Random Forest proves to be marginally more robust due to the significant multicollinearity among features, the more critical finding is that neither model achieves a positive R^2 value. This indicates that the feature set does not effectively predict next-day returns, which aligns with market efficiency theory rather than being a limitation of either model's implementation.

The evidence gathered regarding correlation, multicollinearity, and feature importance across both notebooks—including the detailed analyses of the Relative Strength Index (RSI) and Exponential Moving Average (EMA)—suggests specific, actionable next steps rather than signaling a dead end.

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